

$$31) \quad 2^{50} \text{ ha } 7$$

$$2 \times 2 = 4$$

$$2 \times 2 \times 2 = 8 \equiv_7 1$$

$$2^{50} = \underbrace{2 \times 2 \times 2}_1 \times \underbrace{2 \times 2 \times 2}_{\wedge} \times \dots$$

$$2^{50} = \underbrace{2^3 \times 2^3 \times 2^3 \times \dots \times 2^3}_{16} \times 2^2$$

$$\begin{array}{r} 50 \overline{) 3} \\ 20 \quad 16 \\ \underline{2} \end{array}$$

$$2^{50} = 2^{16 \times 3 + 2} = (2^3)^{16} \times 2^2 \equiv_7 1^{16} \times 4 = 4$$

$$41^{63} = \underbrace{41 \times 41 \times 41 \times \dots \times 41}_{63} \equiv_7 \underbrace{(-1) \cdot (-1) \times \dots \times (-1)}_{63}$$

$$41 \equiv_7 6 \equiv_7 -1$$

$$= -1 \equiv_7 6$$

$$41 \equiv_7 -1 \Rightarrow 41^{63} \equiv (-1)^{63} = -1 \equiv_7 6$$

$$32) \quad 4^{215} \text{ ha } 9$$

$$4 \times 4 = 16 \equiv_9 7 \equiv_9 -2$$

$$4^3 = 64 \equiv_9 1$$

$\xrightarrow{-7 \times 9}$

$$\begin{array}{r} 9 \overline{) 3} \\ 09 \quad 71 \\ \underline{2} \end{array}$$

$$215 = 71 \times 3 + 2$$

$$4^{215} = 4^{71 \cdot 3 + 2} = 4^{71 \cdot 3} \times 4^2$$

$$= (4^3)^{71} \times 4^2 \equiv_9 1^{71} \times 16 = 16 \equiv_9 7$$

daí resto 7

OUTRA RESOLUÇÃO

$$4^{215} = (2^2)^{215} = 2^{430}$$

$$2^3 = 8 \equiv_9 -1$$

$$\begin{array}{r} 4 \overline{) 30} \quad \underline{3} \\ 13 \quad 143 \\ 10 \\ 1 \end{array}$$

$$4^{215} = 2^{430} = 2^{143 \cdot 3} \times 2^1$$

$$= (2^3)^{143} \times 2 \equiv_9 (-1)^{143} \times 2 = -2 \equiv_9 7$$

Exercício que não está na folha 

Resto de $32^{40} + 12^{32}$ por 7

$$32 \equiv_7 4 \Rightarrow 32^{40} \equiv_7 4^{40} =$$

$$4^2 = 16 \equiv_7 2$$

$$4^3 = 64 \equiv_7 1$$

$$\begin{array}{r} 40 \overline{) 3} \\ 10 \quad 13 \\ 1 \end{array}$$

$$4^{40} = (4^3)^{13} \times 4^1 \equiv_7 1^{13} \times 4 = 4$$

ou

$$4^{40} = (4^2)^{20} \equiv_7 2^{20} = 2^{6 \times 3} \times 2^2 = (2^3)^6 \times 4 \equiv_7 1^6 \times 4 = 4$$

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$$12 \equiv_7 5 \equiv_7 -2$$

$$12^{32} \equiv (-2)^{32} = 2^{32} = (2^3)^{10} \times 2^2 \equiv_7 1^{10} \times 2^2 = 4$$

$$\text{Logo } 32^{40} + 12^{32} \equiv_7 4 + 4 = 8 \equiv_7 1$$

daí resto 1

$$33) \quad 11^{10} = (11^5)^2 = (121)^5 \equiv_{100} 21^5 = 21^2 \times 21^2 \times 21 =$$

$$441 \times 441 \times 21 \equiv_{100} 41 \times 41 \times 21 =$$

$$11^2 = 121 \equiv_{100} 21$$

$$1681 \times 21 \equiv_{100} 81 \times 21 = 1701 \equiv_{100} 1$$

$$\begin{array}{r} 21 \\ 21 \\ \hline 21 \end{array}$$

$$\begin{array}{r} 41 \\ 41 \\ \hline 41 \end{array}$$

$$\begin{array}{r} 92 \\ \hline \end{array}$$

$$\begin{array}{r} 164 \\ \hline \end{array}$$

$$441 \equiv_{100} 41$$

$$1681 \equiv_{100} 81$$

$$34) \quad \forall_n \in \mathbb{N} \quad m^3 - m \text{ é múltiplo de } 3$$

$$m^3 - m \equiv_3 0$$

1º Caso

$$m \equiv_3 0 \quad \Rightarrow \quad m^3 - m \equiv_3 0^3 - 0 = 0$$

2º Caso

$$m \equiv_3 1 \quad \Rightarrow \quad m^3 - m \equiv_3 1^3 - 1 = 0$$

3º Caso

$$m \equiv_3 2 \quad \Rightarrow \quad m^3 - m \equiv_3 2^3 - 2 = 6 \equiv_3 0$$

$$35) \quad a \equiv_n b \xRightarrow{\iff} a^2 \equiv b^2$$

		$C_1 \times C_2$	C_2^2
a	a^2	a^3	a^4
$a \equiv_{10} 0$	$a^2 \equiv_{10} 0 = 0$	$a^3 \equiv_{10} 0 \times 0 = 0$	$a^4 \equiv_{10} 0^2 = 0$
$a \equiv_{10} 1$	$a^2 \equiv_{10} 1^2 = 1$	$a^3 \equiv_{10} 1 \times 1 = 1$	$a^4 \equiv_{10} 1^2 = 1$
$a \equiv_{10} 2$	$a^2 \equiv_{10} 2^2 = 4$	$a^3 \equiv_{10} 2 \times 4 = 8$	$a^4 \equiv_{10} 4^2 \equiv_{10} 6$
$a \equiv_{10} 3$	$a^2 \equiv_{10} 3^2 = 9$	$a^3 \equiv_{10} 3 \times 9 \equiv_{10} 7$	$a^4 \equiv_{10} 9^2 \equiv_{10} 1$
$a \equiv_{10} 4$	$a^2 \equiv_{10} 4^2 = 16 \equiv_{10} 6$	$a^3 \equiv_{10} 4 \times 6 \equiv_{10} 4$	$a^4 \equiv_{10} 6^2 \equiv_{10} 6$
$a \equiv_{10} 5$	$a^2 \equiv_{10} 5^2 \equiv_{10} 5$	$a^3 \equiv_{10} 5^2 \equiv_{10} 5$	$a^4 \equiv_{10} 5^2 \equiv_{10} 5$
$a \equiv_{10} 6$	$a^2 \equiv_{10} 6^2 \equiv_{10} 6$	$a^3 \equiv_{10} 6^2 \equiv_{10} 6$	$a^4 \equiv_{10} 6^2 \equiv_{10} 6$
$a \equiv_{10} 7$	$a^2 \equiv_{10} 7^2 \equiv_{10} 9$	$a^3 \equiv_{10} 7 \times 9 \equiv_{10} 3$	$a^4 \equiv_{10} 9^2 \equiv_{10} 1$
$a \equiv_{10} 8$	$a^2 \equiv_{10} 8^2 \equiv_{10} 4$	$a^3 \equiv_{10} 8 \times 4 \equiv_{10} 2$	$a^4 \equiv_{10} 4^2 \equiv_{10} 6$
$a \equiv_{10} 9$	$a^2 \equiv_{10} 9^2 \equiv_{10} 1$	$a^3 \equiv_{10} 9 \times 1 = 9$	$a^4 \equiv_{10} 1^2 = 1$

$$36) \quad \overline{3x5y} \equiv_4 0 \iff \overline{5y} \equiv_4 0 \Rightarrow y=2 \vee y=6$$

$$\text{Se } y=2$$

$$\overline{3n52} \equiv_q 0 \Leftrightarrow$$

$$3+n+5+2 \equiv_q 0 \Leftrightarrow$$

$$n+10 \equiv_q 0 \Leftrightarrow$$

$$n+1 \equiv_q 0 \Rightarrow n=8 \quad 3852$$

$$\text{Se } y=6$$

$$\overline{3n56} \equiv_q 0 \Leftrightarrow 3+n+5+6 \equiv_q 0 \Leftrightarrow n+14 \equiv_q 0$$

$$\Leftrightarrow n+5 \equiv_q 0 \Rightarrow n=4 \quad 3456$$

$$38) \quad a - 25n \equiv_{29} 15$$

$$\text{m.d.c}(25, 29) = 1$$

$$25n \equiv_{29} 15 \Leftrightarrow 5 \equiv_{29} 3 \stackrel{\times 6}{\Leftrightarrow} 30 \equiv_{29} 18$$

$$\Leftrightarrow n \equiv_{29} 18 \Leftrightarrow n = 18 + 29k$$

$$b - 5n \equiv_{26} 2$$

$$\text{m.d.c}(5, 26) = 1 \mid 2$$

$$5n \equiv_{26} 2 \stackrel{\times 5}{\Leftrightarrow} 25n \equiv_{26} 10 \Leftrightarrow -n \equiv_{26} 10$$

$$\stackrel{\times (-1)}{\Leftrightarrow} n \equiv_{26} -10 \Leftrightarrow n = -10 + 26k, \quad k \in \mathbb{Z}$$

$$c - 140n \equiv_{301} 133$$

$$\text{m.d.c}(140, 301) = 7 \nmid 133$$

$$140n \equiv_{301} 133 \stackrel{/7}{\Leftrightarrow} 20n \equiv_{43} 19$$

$$20n \equiv_{43} 19 \iff 300n \equiv_{43} 285$$

$$\iff -n \equiv_{43} 285 \iff n \equiv_{43} -285 \iff n \equiv_{43} 16$$

39) a- $12n \equiv_{16} 6$

$$m.d.c(12, 16) = 4 \nmid 6 \quad \text{não tem solução}$$

b- $12n \equiv_{35} 7 \iff 35 \mid 12n - 7 \iff \exists y \in \mathbb{Z}, 12n - 7 = 35y$

$$m.d.c(12, 35) = 1 \mid 7 \quad \text{tem solução}$$

$$12n \equiv_{35} 7 \xrightarrow[\substack{\times 3 \\ m.d.c(12, 35)=1}]{\iff} 36n \equiv_{35} 21 \iff n \equiv_{35} 21$$

$$\Rightarrow n = 21 + 35K, K \in \mathbb{Z}$$

c- $12n \equiv_{35} 24$

$$m.d.c(12, 35) = 1 \mid 4$$

$$12n \equiv_{35} 24 \xrightarrow{\times 3} 36n \equiv_{35} 72 \iff n \equiv_{35} 2$$

$$n = 2 + 35K, K \in \mathbb{Z}$$

d- $10n \equiv_{16} 14$

$$m.d.c(10, 16) = 2 \mid 14$$

$$10n \equiv_{16} 14 \iff 5n \equiv_8 7 \xrightarrow{\times 5} 25n \equiv_8 35$$

$$\iff n \equiv_8 3 \iff n = 3 + 8K, K \in \mathbb{Z}$$

e- $60n \equiv_{165} -30$

$$60 = 4 \times 15 = 2^2 \times 3 \times 5$$

$$165 = 3 \times 5 \times 11$$

$$\text{m.d.c.}(60, 165) = 15 \mid -30$$

$$4x \equiv_{11} -2 \Leftrightarrow 12x \equiv_{11} -6 \Leftrightarrow x \equiv_{11} -6$$

$$\Leftrightarrow x \equiv 5 \Leftrightarrow x = 5 + 11K, K \in \mathbb{Z}$$

$$41) 13x \equiv_{42} 17$$

$$\text{m.d.c.}(13, 42) = 1 \mid 17$$

$$42 = 2 \times 3 \times 7 = 6 \times 7$$

$$13x \equiv_{42} 17 \quad \begin{cases} 13x \equiv_6 17 \\ 13x \equiv_7 17 \end{cases} \quad (\Leftrightarrow) \quad \begin{cases} x \equiv_6 -1 \\ -x \equiv_7 17 \end{cases}$$

$$\Leftrightarrow \begin{cases} x \equiv_6 -1 \\ -x \equiv_7 -4 \end{cases} \quad (\Leftrightarrow) \quad \begin{cases} x \equiv_6 -1 \\ x \equiv_7 4 \end{cases} \quad (\Leftrightarrow) \quad \begin{cases} 4 + 7K \equiv_6 -1 \\ x = 4 + 7K \end{cases}$$

$$\Leftrightarrow \begin{cases} 7K \equiv_6 -5 \\ - \end{cases} \quad \Leftrightarrow \begin{cases} K \equiv_6 1 \\ - \end{cases} \quad \Leftrightarrow \begin{cases} K = 1 + 6t \\ - \end{cases}$$

$$x = 4 + 7K$$

$$\Leftrightarrow x = 4 + 7(1 + 6t)$$

$$\Leftrightarrow x = 11 + 42t, t \in \mathbb{Z}$$